

Problem Statement

South Africa + Sri Lanka Joint Agricultural Small Language Model (AgriSLM)

Introduction

Agriculture plays an important role in the economy and in the sustenance of people in Sri Lanka (SL) and South Africa (SA). Crop yield depends on timely response to changing climate-related issues and soil conditions. This AgriSLM is a prototype toward a local language model to be implemented in Phase 2.

This model combines two distinct regional variations from SL and SA. It is a collaborative building of a domain-specific agricultural Small Language Model (SLM) that supports:

- Climate-aware reasoning
- Crop advisory for both countries
- Offline rural deployment
- A clean, reproducible pipeline suitable for TRI evaluation

Rationale

Sri Lanka Variant

The problem of **limited access to scientific, timely, and localized agricultural guidance for economically vulnerable smallholder farmers** has been a long-standing issue.

The SL variant focuses on the Nuwara Eliya region, where the majority of the Tamil tea estate plantation community lives. They were brought to Sri Lanka from South India by the British 200 years ago to transform the jungle into a tea industry. Even today, they remain among the most vulnerable and landless communities in Sri Lanka.

“Despite positive law reform, Hill Country Tamils remain one of the most discriminated against and economically, socially and politically marginalized communities in the country. By many developmental indicators, this community is the ‘furthest behind’ in Sri Lanka.” (*Verité Research, 2019*)

As stated above, the farmers are economically poor. They do not own land, which compels them to lease land for cultivation. This forces them to take loans from local lenders, as they lack collateral for formal banking. There are no insurance schemes protecting these farmers from crop failure.

Domination by intermediaries depresses farm-gate prices, meaning farmers often cannot recoup basic production costs. This renders a poor farmer in Nuwara Eliya completely vulnerable.

Extreme weather, climate change, soil degradation, and severe economic constraints pose serious challenges to vegetable cultivation in the region.

Farming methods are traditional, and nonscientific practices contribute to soil erosion and landslides. Most agricultural resources do not reach these farmers due to the lack of materials in local languages. This is a critical gap that must be narrowed.

South Africa Variant.

The purpose of this research is to classify the historical and contemporary challenges affecting local African farmers in South Africa, particularly those who remain dependent on small household plots, communal land, or leased land instead of inheriting productive agricultural land capable of sustaining commercial farming operations.

Limited access to fertile land, water resources, agricultural finance, and modern farming infrastructure continues to constrain the growth of many emerging farmers and rural communities.

The roots of these challenges can be traced to the **Natives Land Act (Act No. 27 of 1913)**, enacted on **19 June 1913**, which restricted Black South Africans from owning or leasing land outside designated reserves.

This legislation, together with the **Native Trust and Land Act of 1936** and subsequent apartheid policies, institutionalised unequal land ownership, fragmented rural communities, and reduced opportunities for intergenerational agricultural development. These historical policies continue to influence South Africa's agricultural landscape.

Since the establishment of democracy in **1994**, the government has pursued land reform through restitution, redistribution, and tenure reform to address historical dispossession while maintaining agricultural productivity and food security.

The government should consider adopting modern policy frameworks to strengthen the Preservation and Development of Agricultural Land Act and introduce fair legislation that supports local farmers. These policies should recognize and encourage sustainable farming practices, innovation, and rural agricultural development.

By embracing proven agricultural solutions and technologies from around the world while adapting them to local needs, the country can improve productivity, strengthen food security, and stimulate long-term economic growth.

The Current Solutions

The current solutions involve **traditional knowledge, informal advice networks, and reactive decision-making**, which pose several challenges.

In the absence of a robust response system and limited access to scientific farming approaches, farmers rely on impromptu decisions. While traditional knowledge is valuable, without scientific support it exposes farmers to high risk. When crops fail, the poor farmer bears the entire burden.

The Proposed Model

This research aims to **develop a domain-specific agricultural Small Language Model that provides timely, localized, and scientifically grounded guidance to smallholder farmers**, enabling them to make informed decisions and reduce vulnerability to climate and economic shocks.

Objectives

1. Provide timely information for effective planning

The model will be trained to give accurate weather forecasts and patterns, **crop diversity in the region**, pesticide guidance, fertilizing timetables, and other agronomic advisories. This will empower farmers to plan effectively for better yield. The statistics provided will assist farmers in making informed crop choices for each season.

2. Provide a robust response system to unforeseen climatic events

The model will answer questions related to unexpected climatic challenges, guiding farmers toward appropriate field responses and connecting them to relevant support mechanisms.

3. Cultivate scientific farming methods as an antidote to soil erosion and low yield

Successful implementation will help farmers enhance local knowledge with scientific solutions.

4. Create small climate tracker devices to monitor environmental conditions and calculate accurate geographical stats.

The device is holistically integrated with an AI computing mechanism to mimic the climate and environmental focus.

Values-Led Problem Statement

Smallholder farmers in South Africa and Sri Lanka, particularly the landless Tamil tea estate workers and marginalized Black South African communities, face severe economic vulnerability due to a lack of timely, scientific agricultural guidance.

This information gap, rooted in historical discrimination and land dispossession, leaves farmers exposed to devastating climate shocks and soil degradation. The AgriSLM project addresses this by providing an inclusive, localized AI solution that prioritizes fairness and cultural preservation.

By integrating traditional knowledge with climate-aware reasoning in local languages, the model empowers these communities to make informed planning decisions and build resilience against unforeseen environmental events.

Guided by the values of equity and rural accessibility, this initiative aims to bridge the digital divide, ensuring that even the most remote farmers can achieve sustainable yields and long-term economic stability, ultimately lifting them from cycles of poverty and marginalization.

Conclusion

The successful development of this approach will **bridge the gap between vulnerable farmers and reliable scientific agricultural knowledge**, enabling timely decision-making, improving climate resilience, and strengthening economic stability.

It will provide robust answers to unforeseen situations, offer effective guidance for crop planning, and support farmers in overcoming the economic and social challenges they face today.